Software Requirements Specification

for

<Online Food Ordering System>

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This document describes the requirements of the Online Food Ordering System. The system allows users to order food from restaurants through an online platform. It helps in reducing manual effort and saves time for customers. The document includes both functional and non-functional requirements of the system. It also acts as a guide for developers and testers during development.

1.2 Document Conventions

All requirements in this document are written in a structured and organized format. Each section is clearly numbered for easy reference and understanding. Important terms are highlighted and explained properly. Simple and clear language is used throughout the document. This helps all users understand the system requirements easily.

1.3 Intended Audience and Reading Suggestions

This document is intended for developers, testers, and end users of the system. Developers use it to design and build the system effectively. Testers use it to verify whether the system meets the requirements. Customers and administrators use it to understand system functionality. It helps all stakeholders get a clear overview of the system.

1.4 Product Scope

The Online Food Ordering System is designed to simplify food ordering processes. Users can browse menus, select food items, and place orders online. It provides secure payment options and order tracking features. The system improves efficiency and user convenience. It can be used in restaurants, cafes, and food delivery services.

1.5 References

This document follows standard SRS guidelines based on IEEE format. Software Engineering textbooks and study materials are used for reference. Online resources and tutorials related to food ordering systems are also considered. Database and UI design concepts are referred from academic sources. These references ensure accuracy and completeness of the document.

2. Overall Description

2.1 Product Perspective

The Online Food Ordering System is a standalone application connected with restaurant services. It replaces traditional manual food ordering methods. The system interacts with databases to store user and order details. It can be integrated with mobile and web platforms. Future enhancements can include AI-based recommendations and faster delivery tracking.

2.2 Product Functions

The system allows users to register and log in securely. Users can browse restaurant menus and select food items. It provides options to add items to cart and place orders. Payment options such as UPI, card, or cash on delivery are available. The system also allows users to track their orders in real time.

2.3 User Classes and Characteristics

The system includes different types of users such as customers, admin, and delivery staff. Customers use the system to order food and track delivery. Admin manages users, restaurants, and system data. Delivery staff handle order delivery to customers. All users are expected to have basic knowledge of mobile or web applications.

2.4 Operating Environment

The system operates on devices such as smartphones, tablets, and computers. It supports operating systems like Android, iOS, and Windows. The application requires internet connectivity to function properly. A database such as MySQL is used to store data. The system runs on standard hardware with no special requirements.

2.5 Design and Implementation Constraints

The system requires a stable internet connection for proper functioning. Payment gateway integration must follow security standards. Database design should ensure fast data access and storage. The system must be compatible with different devices and platforms. Security and performance must be maintained during development.

2.6 User Documentation

User manuals will be provided to guide users in using the system. Help sections will be available within the application. Instructions will be simple and easy to understand. Training materials may be provided if required. Documentation will be updated regularly based on system changes.

2.7 Assumptions and Dependencies

It is assumed that users have internet access and basic knowledge of using apps. Restaurants are responsible for updating their menus regularly. Payment gateways are

expected to function properly. The system depends on server availability for smooth operation. Future updates may depend on user feedback.

3. External Interface Requirements

3.1 User Interfaces

The system provides a simple and user-friendly interface. It includes login pages, dashboards, and order screens. Navigation is easy with clearly labeled buttons and menus. The design is attractive and responsive for all devices. Error messages are displayed clearly for better understanding.

3.2 Hardware Interfaces

The system requires devices such as smartphones or computers. It may also use GPS for location-based services. Hardware should support internet connectivity. No special hardware is required for normal usage. Standard devices are sufficient for system operation.

3.3 Software Interfaces

The system uses a database such as MySQL for storing data. It connects with payment gateways for transactions. The application interacts with operating systems for execution. Different modules communicate internally within the system. All software components must work together smoothly.

3.4 Communication Interfaces

The system uses internet connectivity for communication. Protocols such as HTTP or HTTPS are used for data transfer. Secure communication is maintained to protect user data. The network should support fast and reliable communication. Data exchange between server and user should be efficient.

4. System Features

4.1 User Registration Module

4.1.1 Description and Priority

This module allows new users to create an account in the system. It stores user details securely in the database. It is an important feature for system access. The module ensures that only valid users can access the system. Hence, it is given high priority.

4.1.2 Stimulus/Response Sequences

When a user enters registration details, the system verifies the information. If valid, the account is created successfully. If invalid, an error message is displayed. The system stores user data securely. Confirmation is provided to the user.

4.1.3 Functional Requirements

The system shall allow usersto register. It shall validate user inputs. It shall store user data securely. It shall prevent duplicate accounts. It shall display confirmation messages.

4.2 Order Processing Module

4.2.1 Description and Priority

This module handles food ordering and processing. It allows users to select items and place orders. It ensures accurate order management. The module is essential for system functionality. It is assigned high priority.

4.2.2 Stimulus/Response Sequences

User selects food items and adds them to the cart. The system calculates total price. User confirms the order and selects payment method. The system processes the order and generates order ID. Confirmation is shown to the user.

4.2.3 Functional Requirements

The system shall allow adding items to cart. It shall calculate total cost. It shall process orders correctly. It shall generate order ID. It shall store order details.

5. Other Non-Functional Requirements

5.1 Performance Requirements

The system should respond quickly to user actions. It should handle multiple users at the same time. Database operations must be efficient. The system should not crash during peak usage. Performance should remain stable under normal conditions.

5.2 Safety Requirements

The system should prevent data loss through backups. Errors should be handled properly without affecting performance. Incorrect orders should be minimized. System failures should be managed safely. Safety measures should be implemented throughout.

5.3 Security Requirements

User authentication is required for system access. Data should be stored securely in the database. Unauthorized access must be prevented. Encryption techniques can be used for protection. User privacy should be maintained at all times.

5.4 Software Quality Attributes

The system should be reliable and easy to use. It must be maintainable and scalable. Performance and accuracy should be high. The system should be flexible for future updates. User satisfaction is an important factor.

5.5 Business Rules

Orders can be placed only during restaurant working hours. Payments must be completed before order confirmation. Admin has full control over the system. Discounts and offers may be applied. System should follow business policies.

6. Other Requirements

The system should support future improvements and updates. Mobile app features can be enhanced further. Integration with AI can improve recommendations. Data backup and recovery features should be included. The system should be scalable for future growth.

Appendix A: Glossary

Customer refers to a user who orders food. Admin manages system operations. Order refers to selected food items. Database stores system data. Payment gateway processes transactions.

Appendix B: Analysis Models

DFD diagrams represent data flow in the system. Use case diagrams show user interactions. ER diagrams define database structure. Flowcharts explain process flow. These models help in understanding the system.

Appendix C: To Be Determined

Some features like AI recommendations are not finalized yet. Advanced delivery tracking may be added later. Cloud integration is under consideration. Some requirements may change in the future. These will be updated later.